Chapter 2
Vector Analysis

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Contents

2.1 Vector Algebra

1. Scalars and Vectors

Scalar: **A quantity has only magnitude (Length, area, volume, temperature, density, energy…).**

Vector: **A quantity has both magnitude and direction (Force, displacement, velocity, acceleration, electric field intensity, magnetic field intensity…).**

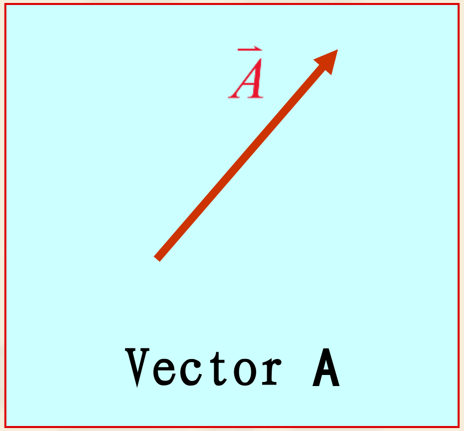
Fields: **The distribution of a quantity in space will constitute a field. Hence, there are scalar fields and vector fields.**

1. Scalars and Vectors

Graphical representation of Vector :

A segment of a directed straight-line.

Algebraic representation ： $\vec{A} = \vec{e}_A A = \vec{e}_A \left|\vec{A}\right|$



Magnitude of Vector : $A = \left|\vec{A}\right|$

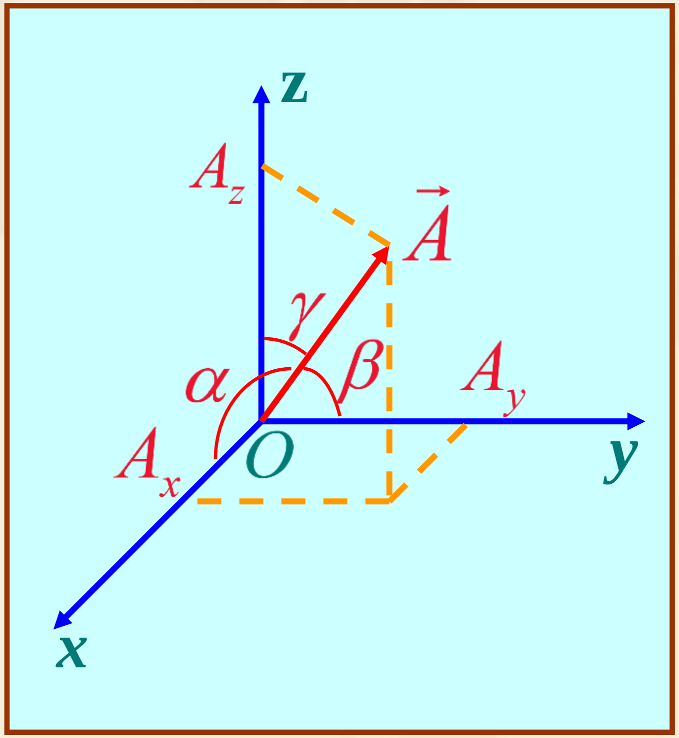
Unit Vector : $\vec{e}_A = \dfrac{\vec{A}}{A}$

Constant Vector : with constant magnitude and fixed direction.

Thinking: Unit Vector = Constant Vector ?

Vector in Cartesian Coordinates

$$\vec{A} = \vec{e}_x A_x + \vec{e}_y A_y + \vec{e}_z A_z$$



$$A_x = A\cos\alpha$$
$$A_y = A\cos\beta$$
$$A_z = A\cos\gamma$$

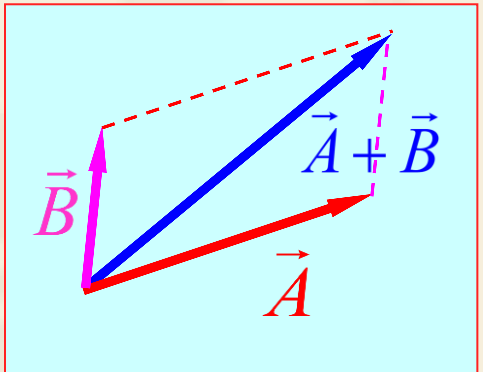
$$\vec{A} = A(\vec{e}_x \cos\alpha + \vec{e}_y \cos\beta + \vec{e}_z \cos\gamma)$$

$$\vec{e}_A = \vec{e}_x \cos\alpha + \vec{e}_y \cos\beta + \vec{e}_z \cos\gamma$$

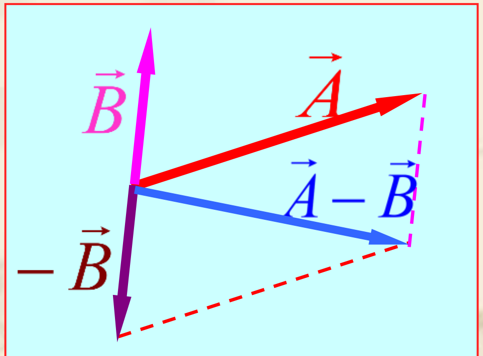
2. Algebraic operation of vectors

（1） Addition and Subtraction of Vectors

$$\vec{A} \pm \vec{B} = \vec{e}_x(A_x \pm B_x) + \vec{e}_y(A_y \pm B_y) + \vec{e}_z(A_z \pm B_z)$$



Addition



Subtraction

Commutative law $\vec{A} + \vec{B} = \vec{B} + \vec{A}$

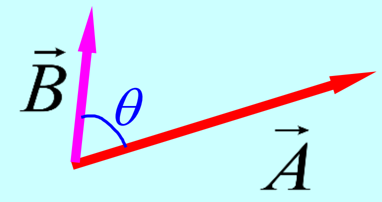
Associative law $\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$

Vector algebra

（2）A vector multiplied by a scalar

$$k\vec{A} = \vec{e}_x kA_x + \vec{e}_y kA_y + \vec{e}_z kA_z$$

（3）Scalar product (dot product)



Angle $\vec{A}$ $\vec{B}$

$$\vec{A} \cdot \vec{B} = AB\cos\theta = A_x B_x + A_y B_y + A_z B_z$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$ ——**Commutative law**

$$\vec{A} \perp \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0 \qquad \vec{A} \parallel \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = AB$$

$$\vec{e}_x \cdot \vec{e}_y = \vec{e}_y \cdot \vec{e}_z = \vec{e}_z \cdot \vec{e}_x = 0$$

$$\vec{e}_x \cdot \vec{e}_x = \vec{e}_y \cdot \vec{e}_y = \vec{e}_z \cdot \vec{e}_z = 1$$

（4）Vector product (cross product)

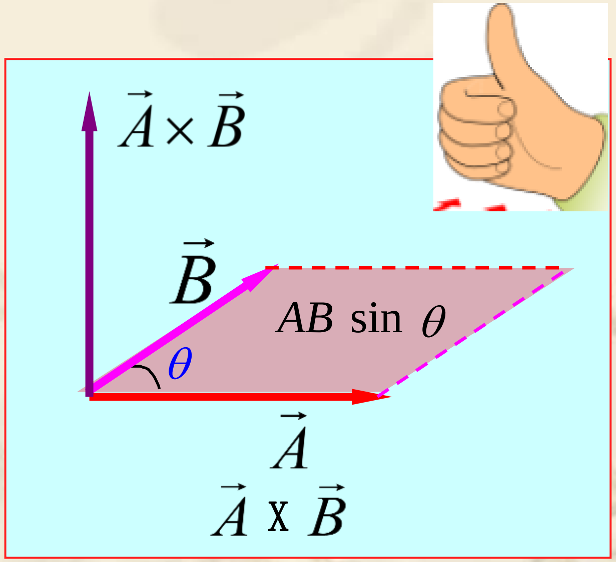
$$\vec{A}\times\vec{B}=\vec{e}_n AB\sin\theta$$

In components:

$$\vec{A}\times\vec{B}=\vec{e}_x(A_yB_z-A_zB_y)+\vec{e}_y(A_zB_x-A_xB_z)+\vec{e}_z(A_xB_y-A_yB_x)$$

In determinant:

$$\vec{A}\times\vec{B}=\begin{vmatrix}\vec{e}_x & \vec{e}_y & \vec{e}_z\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$$



$$\vec{A}\times\vec{B}=-\vec{B}\times\vec{A}$$

If $\vec{A}\,//\,\vec{B}$ **,then** $\left|\vec{A}\times\vec{B}\right|=0$

（5）Vector Operations

$$(\vec{A}+\vec{B})\cdot\vec{C}=\vec{A}\cdot\vec{C}+\vec{B}\cdot\vec{C}$$ —— **Distributive law**

$$(\vec{A}+\vec{B})\times\vec{C}=\vec{A}\times\vec{C}+\vec{B}\times\vec{C}$$ —— **Distributive law**

$$\vec{A}\cdot(\vec{B}\times\vec{C})=\vec{B}\cdot(\vec{C}\times\vec{A})=\vec{C}\cdot(\vec{A}\times\vec{B})$$ —— **Scalar triple product**

$$\vec{A}\times(\vec{B}\times\vec{C})=(\vec{A}\cdot\vec{C})\vec{B}-(\vec{A}\cdot\vec{B})\vec{C}$$ —— **Vector triple product**

Prove them after class

Exercise：

$$\vec{A}=\vec{e}_x 2+\vec{e}_y 3+\vec{e}_z 4$$
$$\vec{B}=\vec{e}_x+\vec{e}_y+\vec{e}_z$$
$$\vec{A}+\vec{B}=$$

1. Result in value

2. Result in graph

$$\vec{A}=\vec{e}_x 2+\vec{e}_y 3+\vec{e}_z 4$$
$$\vec{B}=\vec{e}_x 4+\vec{e}_y 3+\vec{e}_z$$
$$\vec{B}\cdot\vec{A}=$$

$$\vec{A}=\vec{e}_x 2+\vec{e}_y 3+\vec{e}_z 4$$
$$\vec{B}=\vec{e}_x 4+\vec{e}_y 3+\vec{e}_z$$
$$\vec{A}\times\vec{B}=$$

1. Express in three components

2. Express in magnitude and phase

2.2 Orthogonal Coordinate Systems

1. Cartesian Coordinates

Variable x, y, z

Unit vector $\vec{e}_x, \vec{e}_y, \vec{e}_z$

Position $\vec{r} = \vec{e}_x x + \vec{e}_y y + \vec{e}_z z$

Line element $\mathrm{d}\vec{l} = \vec{e}_x \mathrm{d}x + \vec{e}_y \mathrm{d}y + \vec{e}_z \mathrm{d}z$

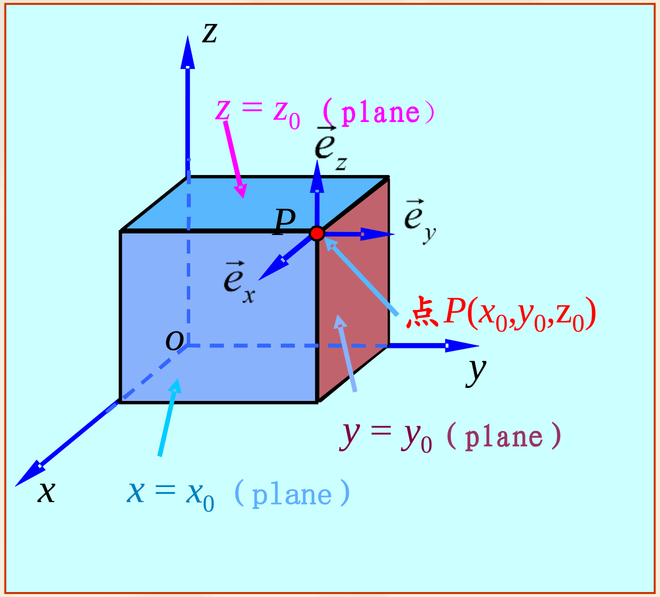
Surface element

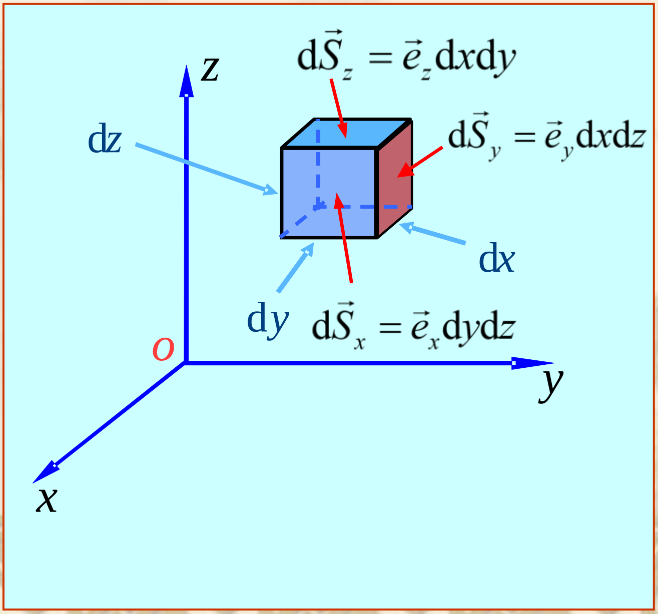
$$\mathrm{d}\vec{S}_x = \vec{e}_x \mathrm{d}l_y \mathrm{d}l_z = \vec{e}_x \mathrm{d}y\mathrm{d}z$$

$$\mathrm{d}\vec{S}_y = \vec{e}_y \mathrm{d}l_x \mathrm{d}l_z = \vec{e}_y \mathrm{d}x\mathrm{d}z$$

$$\mathrm{d}\vec{S}_z = \vec{e}_z \mathrm{d}l_x \mathrm{d}l_y = \vec{e}_z \mathrm{d}x\mathrm{d}y$$

Volume element $\mathrm{d}V = \mathrm{d}x\mathrm{d}y\mathrm{d}z$





2. Cylindrical Coordinates

Variable ρ, ϕ, z

Unit vector $\vec{e}_\rho, \vec{e}_\phi, \vec{e}_z$

Position $\vec{r} = \vec{e}_\rho \rho + \vec{e}_z z$

Line element $\mathrm{d}\vec{l} = \vec{e}_\rho \mathrm{d}\rho + \vec{e}_\phi \rho \mathrm{d}\phi + \vec{e}_z \mathrm{d}z$

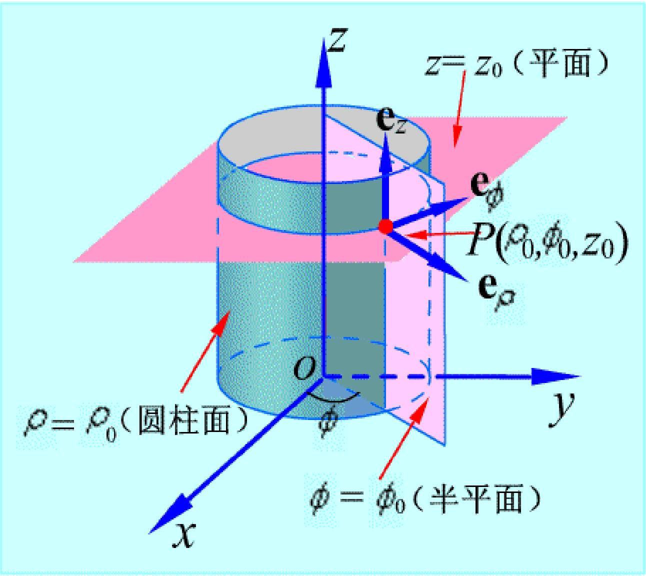
Surface element

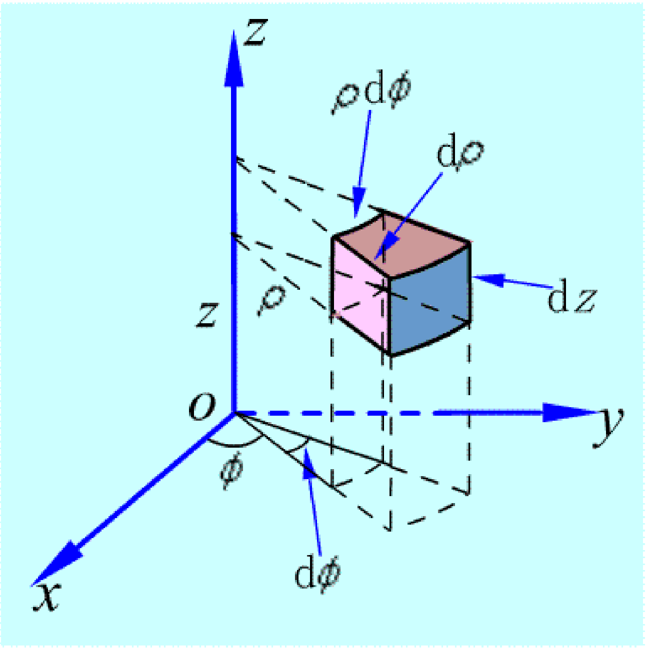
$$\mathrm{d}\vec{S}_\rho = \vec{e}_\rho \mathrm{d}l_\phi \mathrm{d}l_z = \vec{e}_\rho \rho \mathrm{d}\phi \mathrm{d}z$$

$$\mathrm{d}\vec{S}_\phi = \vec{e}_\phi \mathrm{d}l_\rho \mathrm{d}l_z = \vec{e}_\phi \mathrm{d}\rho \mathrm{d}z$$

$$\mathrm{d}\vec{S}_z = \vec{e}_z \mathrm{d}l_\rho \mathrm{d}l_\phi = \vec{e}_z \rho \mathrm{d}\rho \mathrm{d}\phi$$

Volume element $\mathrm{d}V = \rho \mathrm{d}\rho \mathrm{d}\phi \mathrm{d}z$





Operation in cylindrical coordinate systems

$$\vec{A} = \vec{e}_\rho A_\rho + \vec{e}_\phi A_\phi + \vec{e}_z A_z \qquad \vec{B} = \vec{e}_\rho B_\rho + \vec{e}_\phi B_\phi + \vec{e}_z B_z$$

$$\vec{A} \pm \vec{B} = \vec{e}_\rho (A_\rho \pm B_\rho) + \vec{e}_\phi (A_\phi \pm B_\phi) + \vec{e}_z (A_z \pm B_z)$$

$$\vec{A} \bullet \vec{B} = (\vec{e}_\rho A_\rho + \vec{e}_\phi A_\phi + \vec{e}_z A_z) \bullet (\vec{e}_\rho B_\rho + \vec{e}_\phi B_\phi + \vec{e}_z B_z)$$
$$= A_\rho B_\rho + A_\phi B_\phi + A_z B_z$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{e}_\rho & \vec{e}_\phi & \vec{e}_z \\ A_\rho & A_\phi & A_z \\ B_\rho & B_\phi & B_z \end{vmatrix} = \vec{e}_\rho (A_\phi B_z - A_z B_\phi) + \vec{e}_\phi (A_z B_\rho - A_\rho B_z)$$
$$+ \vec{e}_z (A_\rho B_\phi - A_\phi B_\rho)$$

Exercise：

$$\vec{A}=\vec{e}_\rho 2+\vec{e}_\phi 3+\vec{e}_z 4$$
$$k=3$$
$$k\vec{A}=$$

$$\vec{A}=\vec{e}_\rho 2+\vec{e}_\phi 3+\vec{e}_z 4$$
$$\vec{B}=\vec{e}_\rho 4+\vec{e}_\phi 3+\vec{e}_z$$
$$\vec{B}\cdot\vec{A}=$$

$$\vec{A}=\vec{e}_\rho 2+\vec{e}_\phi 3+\vec{e}_z 4$$
$$\vec{B}=\vec{e}_\rho 4+\vec{e}_\phi 3+\vec{e}_z$$
$$\vec{A}\times\vec{B}=$$

E.g. Calculte the integral of function f along the side surface of a cylinder with radius of 1, and height of h, where $f(\vec{r})=\vec{e}_\rho 3\rho z+\vec{e}_z\varphi$

$$\int_s\left(\vec{e}_\rho 3\rho z+\vec{e}_z\varphi\right)\cdot d\vec{s}$$
$$=\int_s\left(\vec{e}_\rho 3\rho z+\vec{e}_z\varphi\right)\cdot\left(\vec{e}_\rho \rho d\varphi dz\right)$$
$$=\int_0^{2\pi}\int_0^{\mathrm{h}}\left(\vec{e}_\rho 3z+\vec{e}_z\varphi\right)\cdot\left(\vec{e}_\rho d\varphi dz\right)$$
$$=\int_0^{2\pi}\int_0^{h}3zd\varphi dz$$
$$=6\pi h$$

3. Spherical Coordinates

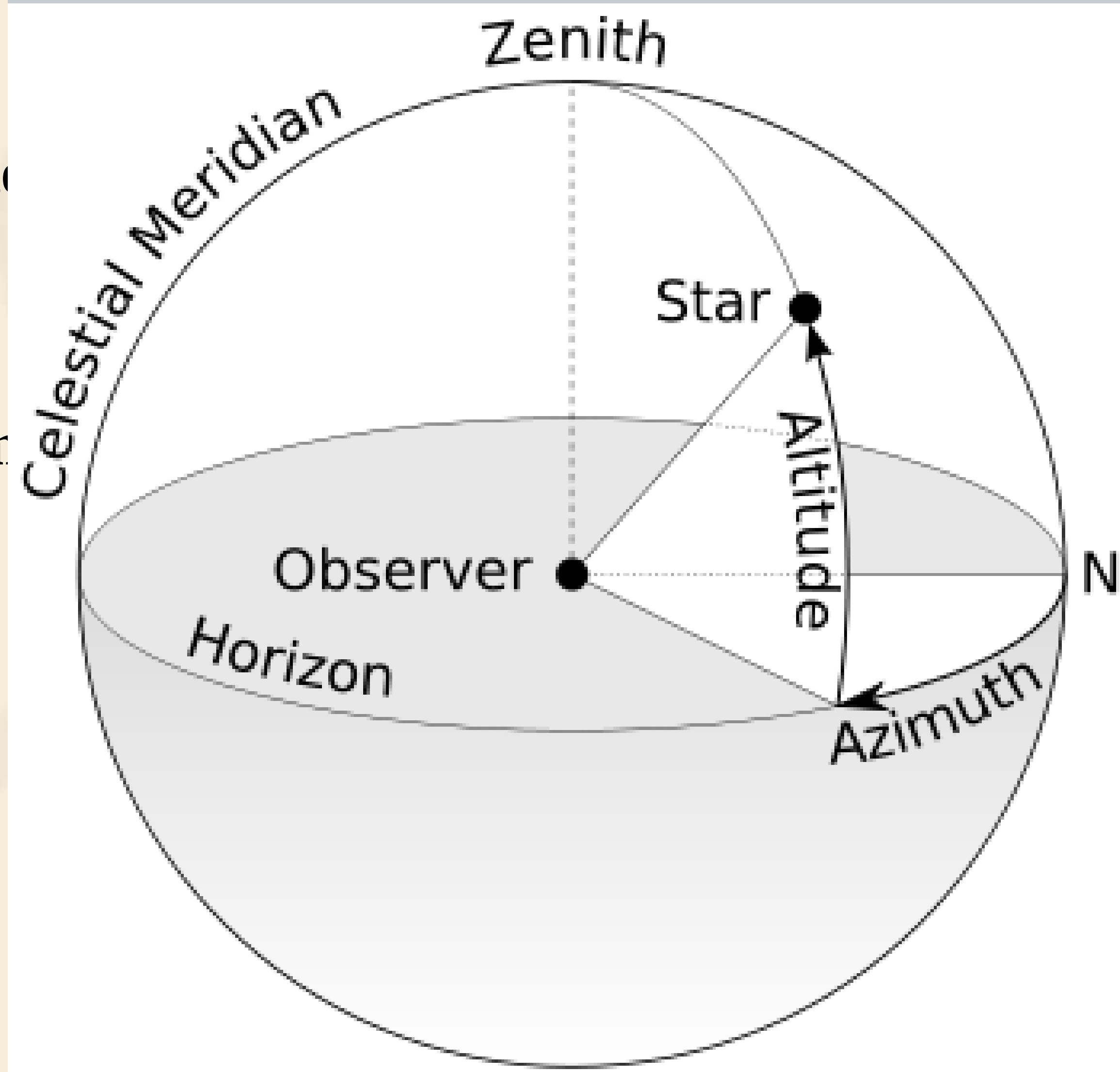
Variable

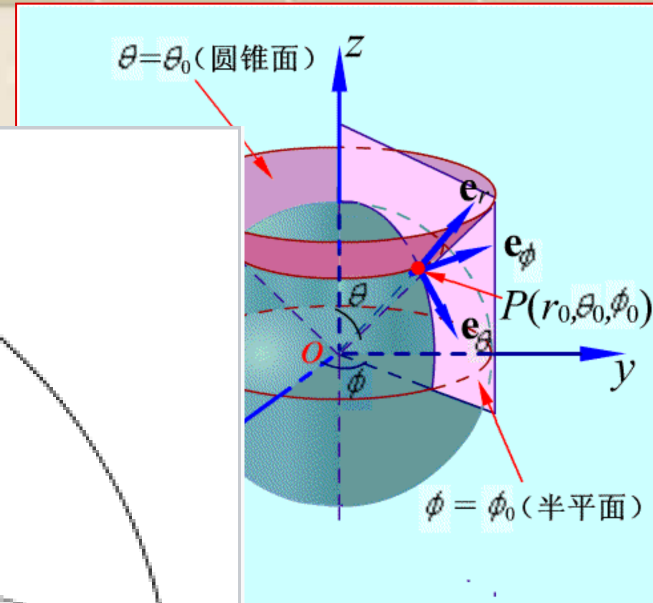
Unit vect

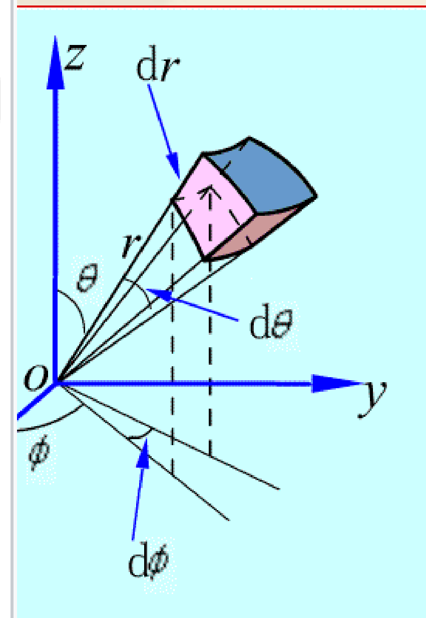
Position

Line elem

Surface element







Volume element

Exercise：

$$\vec{A}=\vec{e}_r 2+\vec{e}_\theta 3+\vec{e}_\phi 4$$
$$k=3$$
$$k\vec{A}=$$

$$\vec{A}=\vec{e}_r 2+\vec{e}_\theta 3+\vec{e}_\phi 4$$
$$\vec{B}=\vec{e}_r 4+\vec{e}_\theta 3+\vec{e}_\phi$$
$$\vec{B}\cdot\vec{A}=$$

$$\vec{A}=\vec{e}_r 2+\vec{e}_\theta 3+\vec{e}_\phi 4$$
$$\vec{B}=\vec{e}_r 4+\vec{e}_\theta 3+\vec{e}_\phi$$
$$\vec{A}\times\vec{B}=$$

E.g. Calculte the integral of function f on the surface of a sphere with radius of 1, where $f(\vec{r})=\vec{e}_r 3r+\vec{e}_\theta 5r$

$$\int_s\left(\vec{e}_r 3r+\vec{e}_\theta 5r\right)\cdot d\vec{s}$$
$$=\int_s\left(\vec{e}_r 3r+\vec{e}_\theta 5r\right)\cdot\left(\vec{e}_r r^2\sin\theta d\varphi d\theta\right)$$
$$=\int_0^{2\pi}\int_0^{\pi}3\sin\theta d\varphi d\theta$$
$$=12\pi$$

4. Relationships

Cartesian and Cylindrical

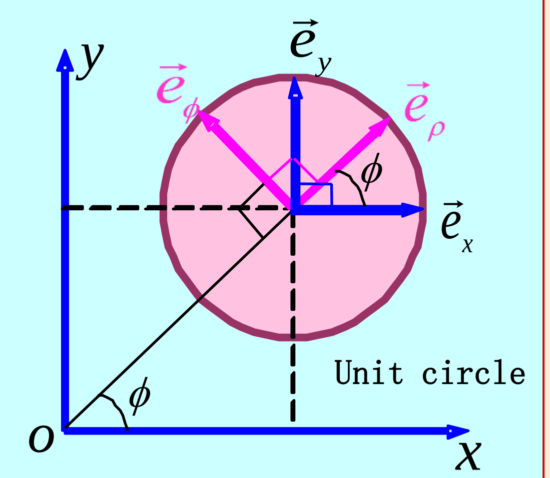
	$\vec{e}_x$	$\vec{e}_y$	$\vec{e}_z$
$\vec{e}_\rho$	$\cos\phi$	$\sin\phi$	0
$\vec{e}_\phi$	$-\sin\phi$	$\cos\phi$	0
$\vec{e}_z$	0	0	1

Cylindrical and Spherical

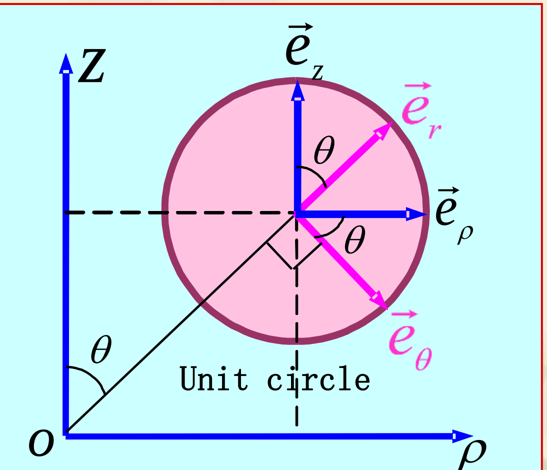
	$\vec{e}_\rho$	$\vec{e}_\phi$	$\vec{e}_z$
$\vec{e}_r$	$\sin\theta$	0	$\cos\theta$
$\vec{e}_\theta$	$\cos\theta$	0	$-\sin\theta$
$\vec{e}_\phi$	0	1	0

Cartesian and Spherical

	$\vec{e}_x$	$\vec{e}_y$	$\vec{e}_z$
$\vec{e}_r$	$\sin\theta\cos\phi$	$\sin\theta\sin\phi$	$\cos\theta$
$\vec{e}_\theta$	$\cos\theta\cos\phi$	$\cos\theta\sin\phi$	$-\sin\theta$
$\vec{e}_\phi$	$-\sin\phi$	$\cos\phi$	0



Cartesian and cylindrical



Cylindrical and spherical

Notes:

- Cartesian system: usually applied in the questions with surface symmetry, e.g., infinitely large surface
- Cylindrical coordinates system: usually applied in the questions with axial symmetry, e.g., the field near to a charged long line.
- Spherical coordinates system: usually applied in the questions with point symmetry, e.g., the field due to a point charge.

2.3 Gradient of a Scalar Field

Scalar field and Vector field

A field is a spatial distribution of a quantity, which may or may not be a function of time.。

- If the quantity is a scalar，it is called scalar field;

  Ex.: Temperature、Potential、Altitude…

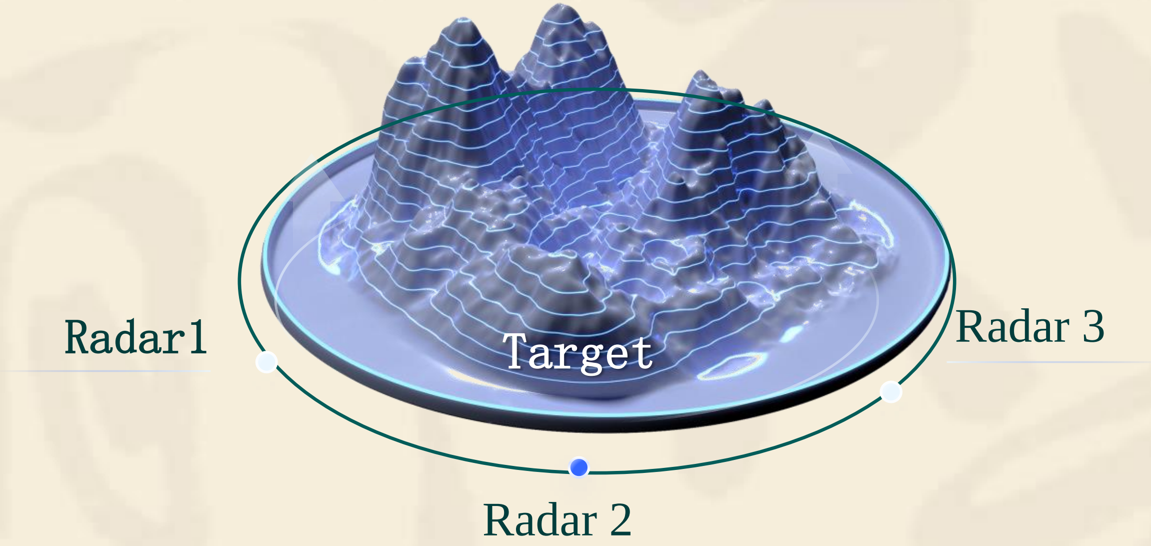
- If the quantity is a vector，it is called vector field;

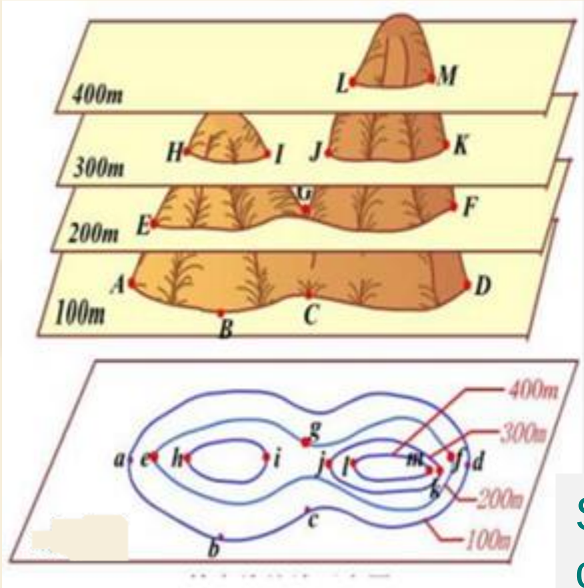
  Ex: Velocity of flow、Gravity、Electric、Magnetic…

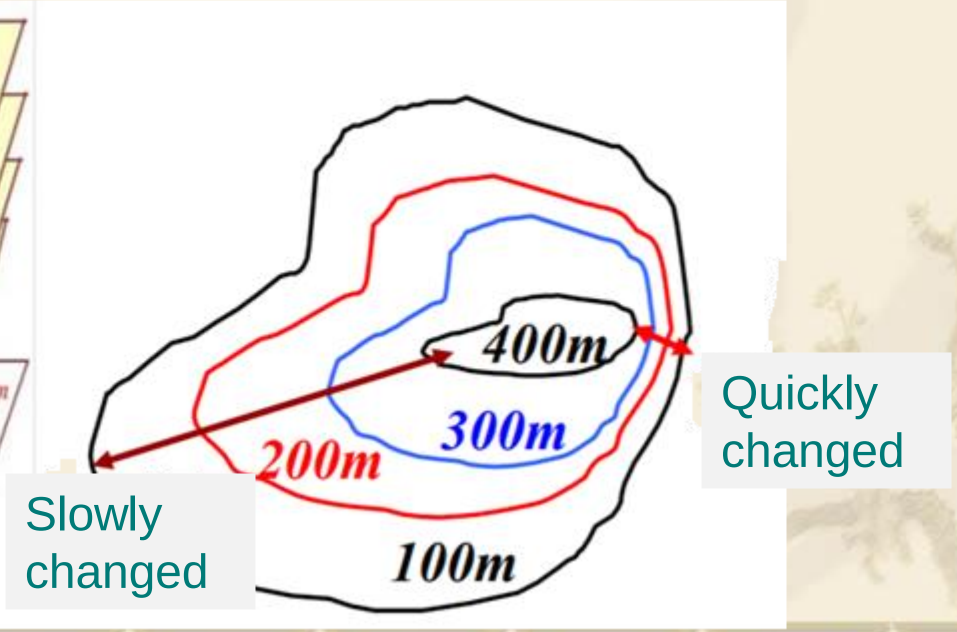
- If it is a function of time, it is called a time-varying field; otherwise it is called a static field.

Expression for static field : $u(x,y,z)$、$\vec{F}(x,y,z)$

Expression for time-varying field : $u(x,y,z,t)$、$\vec{F}(x,y,z,t)$

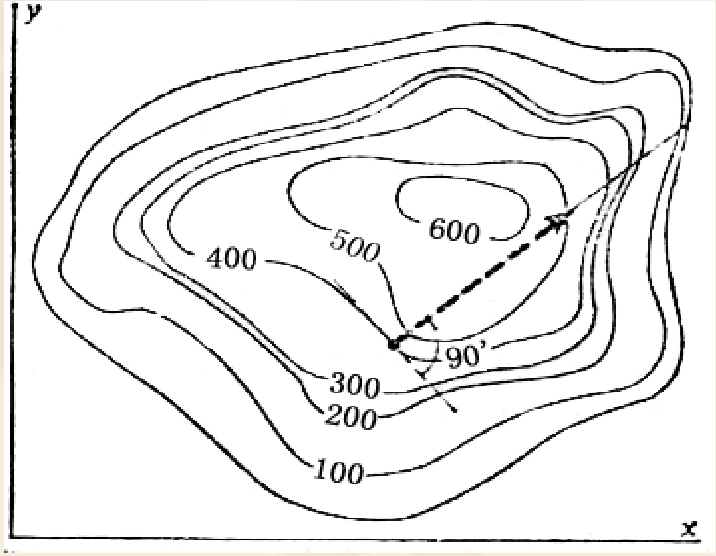

Radar1
Target
Radar 3
Radar 2


400m
L
M
300m
H
I
J
K
200m
E
G
F
100m
A
D
B
C
400m
300m
200m
100m
a
e
h
i
g
j
l
m
f
d
k
c
b


400m
300m
200m
100m
Quickly changed
Slowly changed

1. Level surface

- **Level surface:** **Surface on which the scalar function is equal to a constant;**
- **It describes the distribution of physical quantities in space visually.**

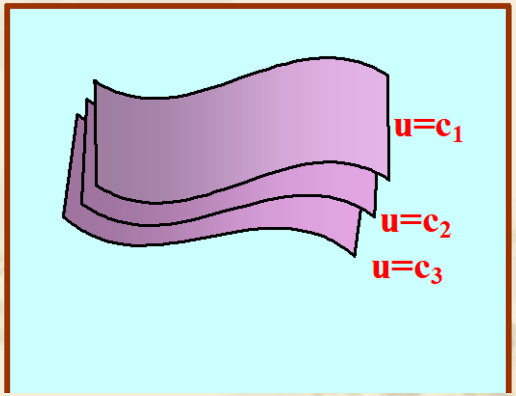


Level surface (line)

Equation : $u(x, y, z) = C$

Characteristics :

- **When the constant C take a series of different values, a series of different level surfaces will be obtained, forming the level surface family;**
- **These surfaces fill up all the space of the scalar field;**
- **These surfaces cannot intersect with each other.**



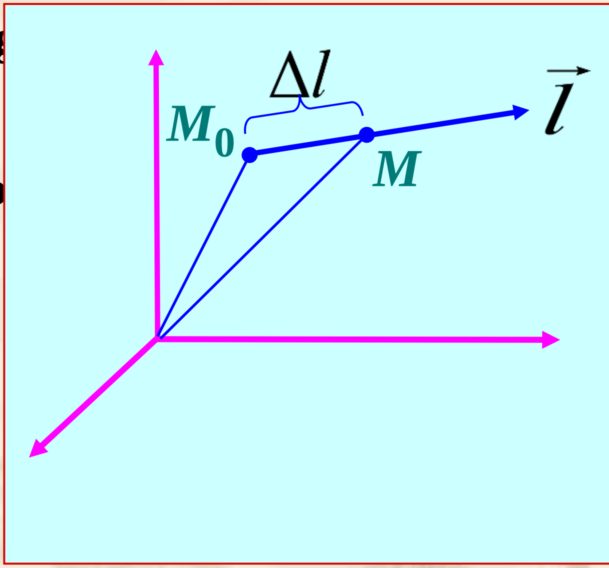
2. Directional derivative

$$\left.\frac{\partial u}{\partial l}\right|_{M_0}=\lim_{\Delta l\to 0}\frac{\Delta u}{\Delta l}=\frac{\partial u}{\partial x}\cos\alpha+\frac{\partial u}{\partial y}\cos\beta+\frac{\partial u}{\partial z}\cos\gamma$$

$\cos\alpha$、 $\cos\beta$、 $\cos\gamma$ —— $\Delta\vec{l}$ **is directional cosines**

Directional derivative: describes the spatial rate of change of a scalar field;

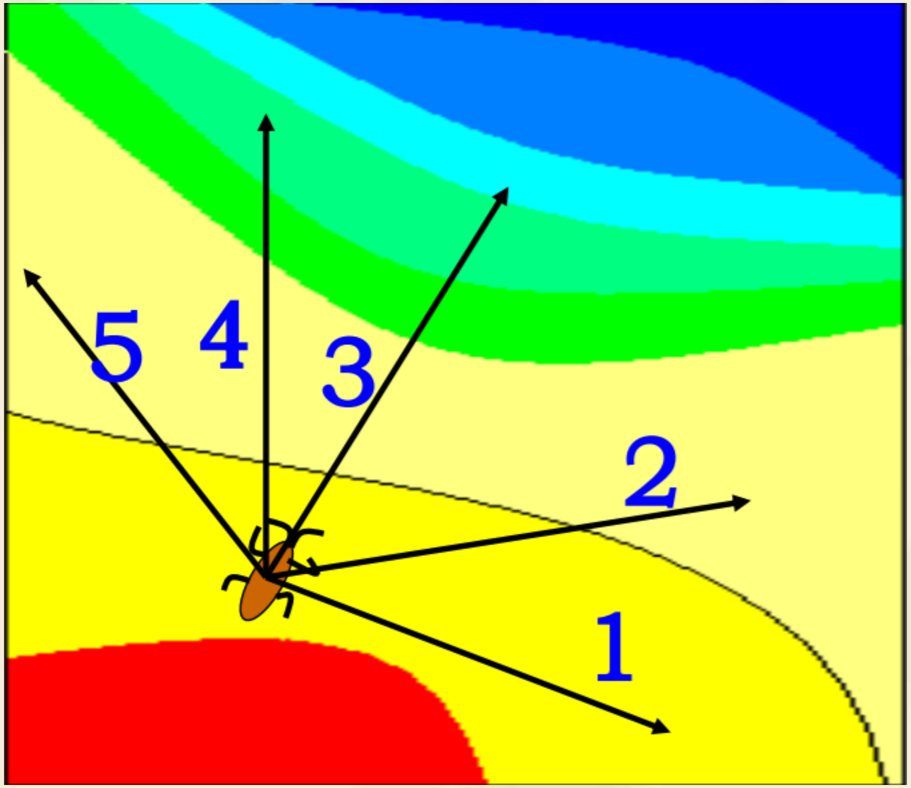
- $\dfrac{\partial u}{\partial l}>0$ —— $u(M)$ **increases along** $\vec{l}$ **line;**
- $\dfrac{\partial u}{\partial l}<0$ —— $u(M)$ **decreases alo**[ng] $\vec{l}$ **line;**
- $\dfrac{\partial u}{\partial l}=0$ —— $u(M)$ **with no change;** $\vec{l}$



Characteristics: **directional derivative is** $\vec{l}$ **related to M_0 as well as line direction.**

Q: What is the maximum spatial rate of change and where?

Q: which direction is the best way to skip?

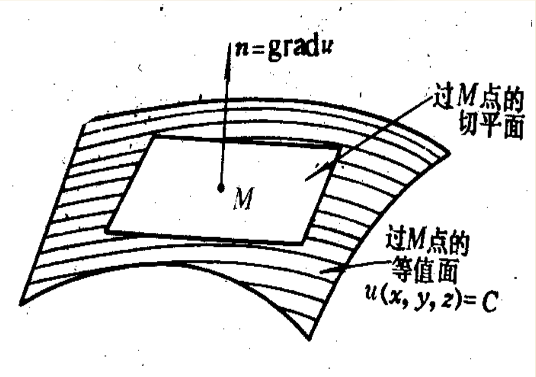


Q: which direction we get the maximum change?

Gradient

3. Gradient of a scalar field （ ∇u or gradu ）

$$\nabla u = \vec{e}_l \frac{\partial u}{\partial l}\bigg|_{\max}$$



- Gradient of a scalar field is a **vector**;
- Describe the **maximum spatial rate of change** of the scalar field at a point, and the direction of maximum value;
- Direction of a gradient at some position is **perpendicular** to the level surface of the scalar field at the position.
- The directional derivative at a position is the projection of the gradient at this position.

Expression of gradient：

Cartesian Coordinates:

$$\nabla u = \vec{e}_x \frac{\partial u}{\partial x} + \vec{e}_y \frac{\partial u}{\partial y} + \vec{e}_z \frac{\partial u}{\partial z}$$

Cylindrical Coordinates:

$$\nabla u = \vec{e}_\rho \frac{\partial u}{\partial \rho} + \vec{e}_\phi \frac{1}{\rho}\frac{\partial u}{\partial \phi} + \vec{e}_z \frac{\partial u}{\partial z}$$

Spherical Coordinates:

$$\nabla u = \vec{e}_r \frac{\partial u}{\partial r} + \vec{e}_\theta \frac{1}{r}\frac{\partial u}{\partial \theta} + \vec{e}_\phi \frac{1}{r\sin\theta}\frac{\partial u}{\partial \phi}$$

Basic equations：

$$\begin{cases} \nabla C = 0 \\ \nabla (Cu) = C\nabla u \\ \nabla (u \pm v) = \nabla u \pm \nabla v \\ \nabla (uv) = u\nabla v + v\nabla u \\ \nabla f(u) = f'(u)\nabla u \end{cases}$$

Example1

Give a scalar $\varphi(x,y,z) = x^2 + y^2 - z$, find:

(1) the gradient of φ at the point $P(1,1,1)$,

and the unit vector of that gradient;

(2) the directional derivative of φ along the unit vector:

$$\vec{e}_l = \vec{e}_x \cos 60^\circ + \vec{e}_y \cos 45^\circ + \vec{e}_z \cos 60^\circ$$

then compare the results and draw a conclusion.

Solution: **(1) from the expression of gradient:**

$$\nabla \varphi\big|_P = [(\vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z})(x^2 + y^2 - z)]_P$$

$$= (\vec{e}_x 2x + \vec{e}_y 2y - \vec{e}_z)\big|_{(1,1,1)} = \vec{e}_x 2 + \vec{e}_y 2 - \vec{e}_z$$

The unit vector of the gradient:

$$\vec{e}_l\big|_P = \frac{\nabla\varphi}{|\nabla\varphi|}\bigg|_P = \frac{\vec{e}_x 2x + \vec{e}_y 2y - \vec{e}_z}{\sqrt{(2x)^2 + (2y)^2 + (-1)^2}}\Bigg|_{(1,1,1)} = \vec{e}_x \frac{2}{3} + \vec{e}_y \frac{2}{3} - \vec{e}_z \frac{1}{3}$$

(2) from the relationship of the directional derivative and gradient, the directional derivative along the e_l:

$$\frac{\partial\varphi}{\partial l} = \nabla\varphi\cdot\vec{e}_l = (\vec{e}_x 2x + \vec{e}_y 2y - \vec{e}_z)\cdot(\vec{e}_x \frac{1}{2} + \vec{e}_y \frac{\sqrt{2}}{2} + \vec{e}_z \frac{1}{2})$$
$$= x + \sqrt{2}y - \frac{1}{2}$$

For a certain point, the value of the directional derivative:

$$\frac{\partial\varphi}{\partial l}\bigg|_P = x + \sqrt{2}y - \frac{1}{2}\bigg|_{(1,1,1)} = \frac{1+2\sqrt{2}}{2}$$

However the gradient of that point is:

$$\left|\nabla\varphi\right|_P = \sqrt{(2x)^2+(2y)^2+(-1)^2}\Big|_{(1,1,1)} = 3$$

Obviously, $\left|\nabla\varphi\right|_P$ **describe the maximum value of** φ **at the point P, the maximum directional derivative, so always is:**

$$\left.\frac{\partial\varphi}{\partial l}\right|_p < \left|\nabla\varphi\right|_p$$

Exercise：

$$f = 2x + 3y^3 + y^2 + 1$$

$$\nabla f =$$

$$f = 2\rho^2 + 2\phi - z$$

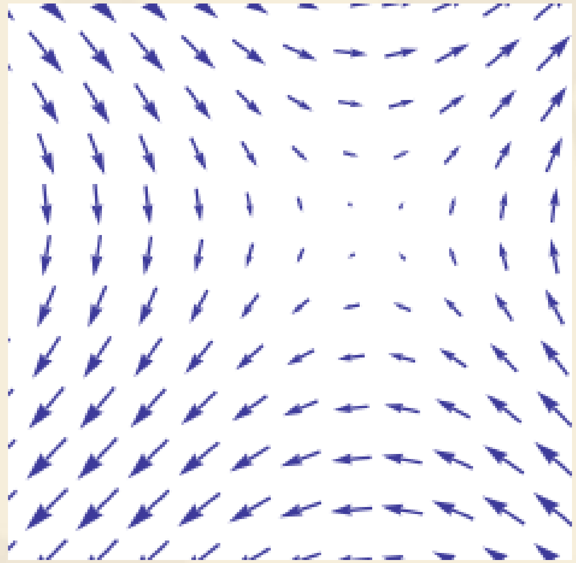
$$\nabla f =$$

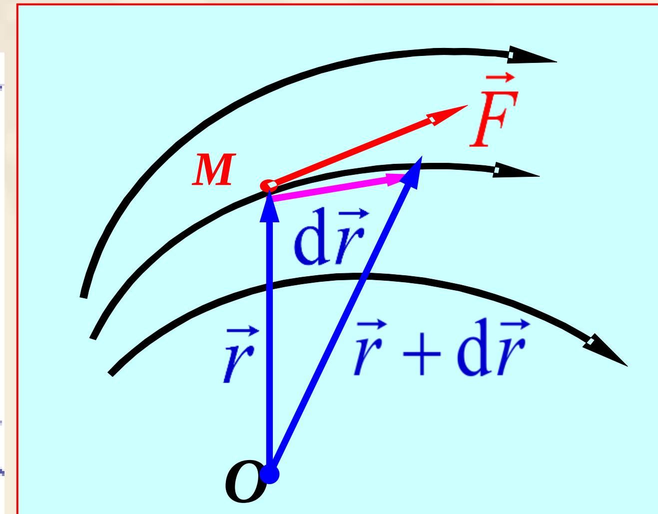
2.4 Flux and divergence of a vector field

Concept： the tangent direction of each point represents the direction of the vector field.

1. Vector line

- **Describe the distribution of the vector field visually**





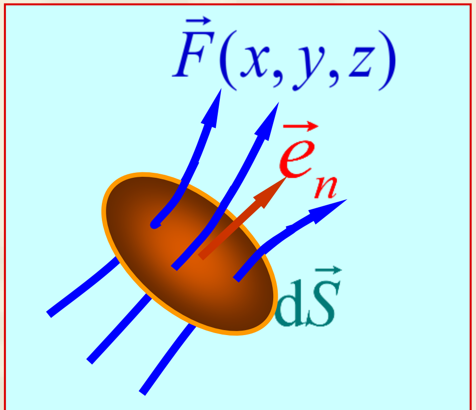
Vector line

Vector line equation：

$$\frac{\mathrm{d}x}{F_x(x,y,z)}=\frac{\mathrm{d}y}{F_y(x,y,z)}=\frac{\mathrm{d}z}{F_z(x,y,z)}$$

2. Flux of a vector field

Q： How to quantitatively describe the size of the vector field？ Flux;



Surface element

$$\psi = \int \mathrm{d}\psi = \int_S \vec{F} \cdot \mathrm{d}\vec{S} = \int_S \vec{F} \cdot \vec{e}_n \mathrm{d}S$$

and: $\mathrm{d}\vec{S} = \vec{e}_n \mathrm{d}S$ —— **surface element;**

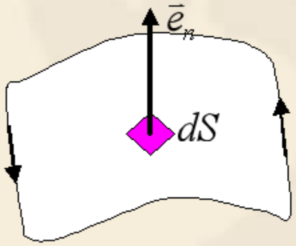
$\vec{e}_n$ ——**The normal unit vector of surface element;**

$\mathrm{d}\psi = \vec{F} \cdot \vec{e}_n \mathrm{d}S$ —— **the flux through ds.**

If the surface S is closed， the direction of S is outward, the flux of the vector field on the closed surface is:

$$\psi = \oint_S \vec{F} \cdot \mathrm{d}\vec{S} = \oint_S \vec{F} \cdot \vec{e}_n \mathrm{d}S$$

About surface element:



1. What is surface element?

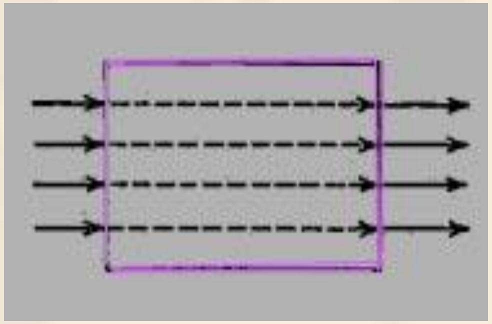
- A surface surrounded by a closed line/curve
- Its area is small
- Its direction is $\vec{e}_n$, it is orthogonal to the area

2. How to decide $\vec{e}_n$?

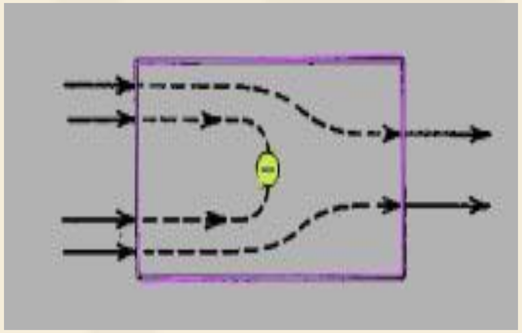
- For un-enclosed surface, it obeys RHR
- For closed surface, outward

- **The physical meaning of flux**

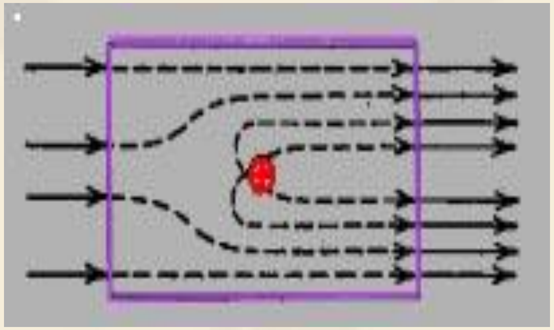
three kinds of possible results:



$\psi = 0$



$\psi < 0$



$\psi > 0$

The vector line are equally in and out

The remained vector line goes in to the closed surface

The remained vector line is out from the closed surface

Flux on the closed surface build the relationship between the flux of a vector field on a closed surface and the source which produce the field.

3. Divergence of a vector field

Definition: the net outward flux of a per unit volume as the volume about the point tends to zero.

$$div\ \vec{F}(x,y,z) = \lim_{\Delta V \to 0} \frac{\oint_S \vec{F}(x,y,z) \cdot \mathrm{d}\vec{S}}{\Delta V}$$

Expression of divergence in Cartesian coordinates

$$div\ \vec{F} = \lim_{\Delta V \to 0} \frac{\oint_S \vec{F} \cdot \mathrm{d}\vec{S}}{\Delta V} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Deduction the expression of divergence in Cartesian coordinates

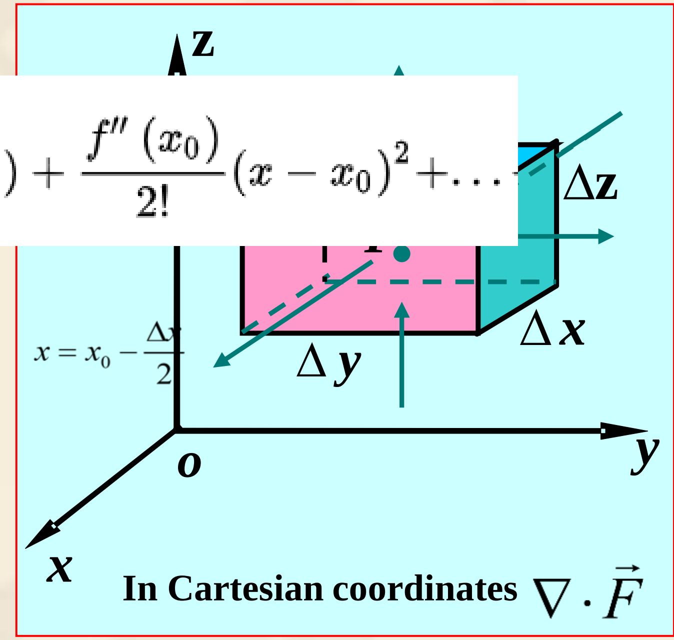
let the volume element ΔV enclosed

point P to be a cube

$F(x_0, y_0, z_0)$. Using t

$$f(x) = \frac{f(x_0)}{0!} + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \ldots$$

series:



$$F_x(x_0 + \frac{\Delta x}{2}, y_0, z_0) \approx F_x(x_0, y_0, z_0) + \frac{\Delta x}{2} \frac{\partial F_x}{\partial x}\Big|_{x_0, y_0, z_0}$$

$$F_x(x_0 - \frac{\Delta x}{2}, y_0, z_0) \approx F_x(x_0, y_0, z_0) - \frac{\Delta x}{2} \frac{\partial F_x}{\partial x}\Big|_{x_0, y_0, z_0}$$

The flux flowing out from the surface elements (both front and back surfaces) $x = x_0 + \frac{\Delta x}{2}$ and $x = x_0 - \frac{\Delta x}{2}$ is:

$$[F_x(x_0 + \frac{\Delta x}{2}, y_0, z_0) - F_x(x_0 - \frac{\Delta x}{2}, y_0, z_0)]\Delta y \Delta z = \frac{\partial F_x}{\partial x} \Delta x \Delta y \Delta z$$

For the same reason， flux through other surface elements could be obtained, then:

$$\oint_S \vec{F}\cdot \mathrm{d}\vec{S} = \frac{\partial F_x}{\partial x}\Delta x\Delta y\Delta z + \frac{\partial F_y}{\partial y}\Delta x\Delta y\Delta z + \frac{\partial F_z}{\partial z}\Delta x\Delta y\Delta z$$

According to the definition of divergence:

$$div\ \vec{F} = \lim_{\Delta V\to 0}\frac{\oint_S \vec{F}\cdot \mathrm{d}\vec{S}}{\Delta V} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$div\ \vec{F} = \nabla \cdot \vec{F}$$

The expression of divergence：

Cartesian Coordinates:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Cylindrical Coordinates:

$$\nabla \cdot \vec{F} = \frac{\partial(\rho F_\rho)}{\rho \partial \rho} + \frac{\partial F_\phi}{\rho \partial \phi} + \frac{\partial F_z}{\partial z}$$

Spherical Coordinates:

$$\nabla \cdot \vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 F_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta F_\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(F_\phi)$$

The related formulas of divergence：

$$\begin{cases} \nabla \cdot \vec{C} = \nabla \cdot \vec{C} = 0 \; (\vec{C} \quad constant) \\ \nabla \cdot (\vec{C} f) = \vec{C} \cdot \nabla f \\ \nabla \cdot (k\vec{F}) = k\nabla \cdot \vec{F} \; (k \quad constant) \\ \nabla \cdot (f\vec{F}) = f\nabla \cdot \vec{F} + \vec{F} \cdot \nabla f \\ \nabla \cdot (\vec{F} \pm \vec{G}) = \nabla \cdot \vec{F} \pm \nabla \cdot \vec{G} \end{cases}$$

Exercise：

$$\vec{A} = \vec{e}_x 2 + \vec{e}_y 3 + \vec{e}_z 4$$

$$\nabla \cdot \vec{A} =$$

$$\vec{A} = \vec{e}_x 2x + \vec{e}_y 3y + \vec{e}_z 4z^2$$

$$\nabla \cdot \vec{A} =$$

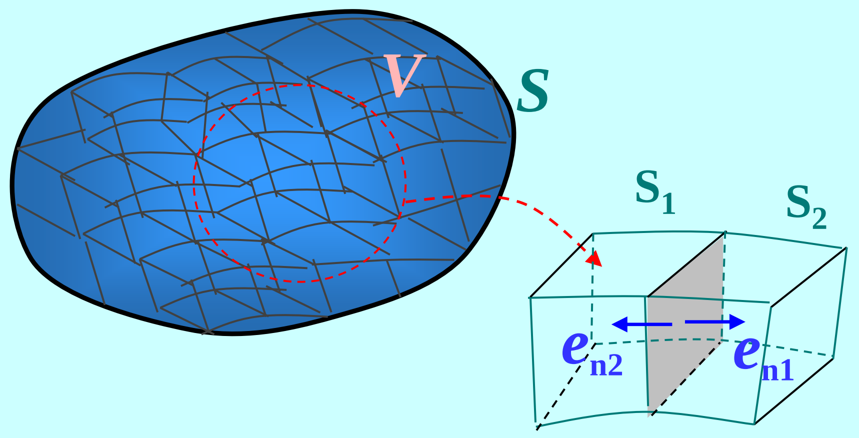
$$\vec{A} = \vec{e}_x 2z^2 + \vec{e}_y 3x$$

$$\nabla \cdot \vec{A} =$$

4. The divergence theorem

The relationship between flux and divergence:

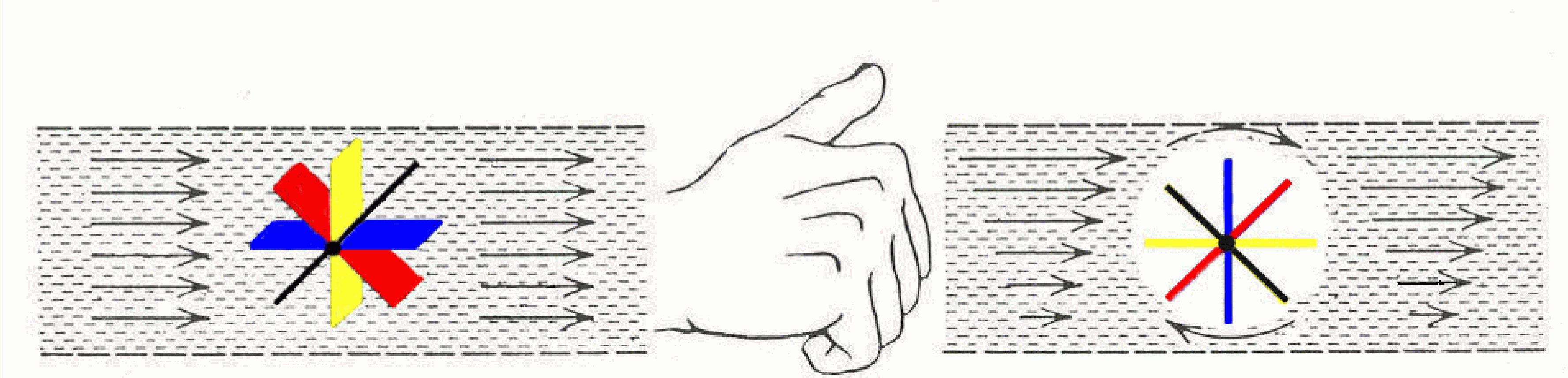
$$\oint_S \vec{F} \cdot \mathrm{d}\vec{S} = \int_V \nabla \cdot \vec{F} \mathrm{d}V$$



Subdivided volume

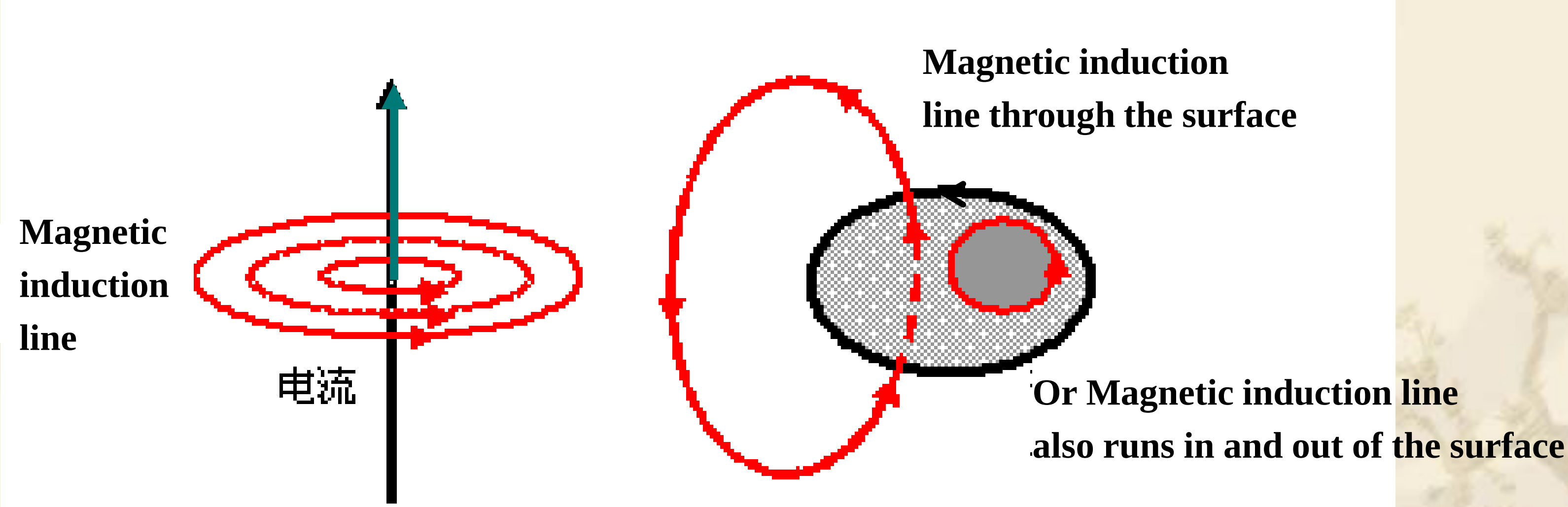
2.5 Circulation and curl of the vector field

1. Circulation and vortex source of the vector field



The relationship between the circulation and the current of the magnetic field:

$$\oint_C \vec{B}(x,y,z)\cdot \mathrm{d}\vec{l} = \mu_0 I = \mu_0 \int_S \vec{J}(x,y,z)\cdot \mathrm{d}\vec{S}$$



The definition of circulation：

the circulation of a vector field around a closed path is defined as the scalar line integral of the vector over the path.

$$\Gamma = \oint_C \vec{F}(x, y, z) \cdot \mathrm{d}\vec{l}$$

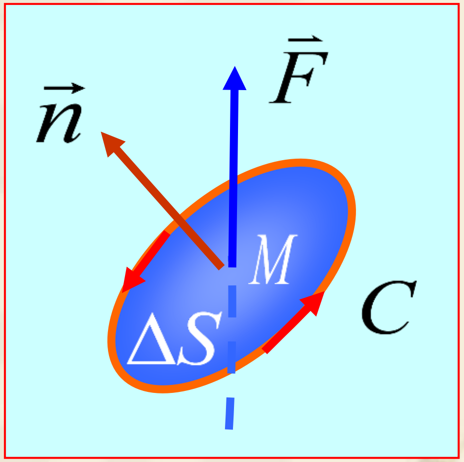
- dl is the tangential direction of the line with much small length.
- If circulation is non-zero, there exists some vortex source on the surface bounded by C。

2. Curl of a vector field（ $\nabla\times\vec{F}$ ）

（1）Circulation area density

Circulation of closed curve around the unit surface element boundary:

$$\mathrm{rot}_n\vec{F}=\lim_{\Delta S\to 0}\frac{1}{\Delta S}\oint_C \vec{F}\cdot \mathrm{d}\vec{l}$$



Known as the circulation area density of vector field at the point M along the direction of $\vec{n}$

Characteristic: **The value is related to the direction of point M.**

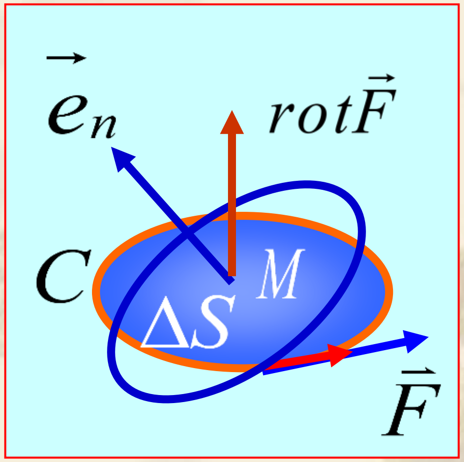
The circulation area density of surface element at any orientation, is the projection of the maximum circulation area density at that point:

$$\mathrm{rot}_n\vec{F} = \vec{e}_n \bullet rot\vec{F}$$

$$= (\vec{e}_x\cos\alpha + \vec{e}_y\cos\beta + \vec{e}_z\cos\gamma) \bullet (\vec{e}_x\mathrm{rot}_x\vec{F} + \vec{e}_y\mathrm{rot}_y\vec{F} + \vec{e}_z\mathrm{rot}_z\vec{F})$$

（2）Curl of a vector field

$$\mathrm{rot}\ \vec{F}$$



- **Expression of $\mathrm{rot}_x\vec{F}$、$\mathrm{rot}_y\vec{F}$ 、$\mathrm{rot}_z\vec{F}$ in Cartesian coordinates**

$\mathrm{rot}_x\vec{F}$ **deduction:**

$$\oint_C \vec{F}\cdot d\vec{l} = \int_{\Delta l_1} \vec{F}\cdot d\vec{l} + \int_{\Delta l_2} \vec{F}\cdot d\vec{l} + \int_{\Delta l_{13}} \vec{F}\cdot d\vec{l} + \int_{\Delta l_4} \vec{F}\cdot d\vec{l}$$

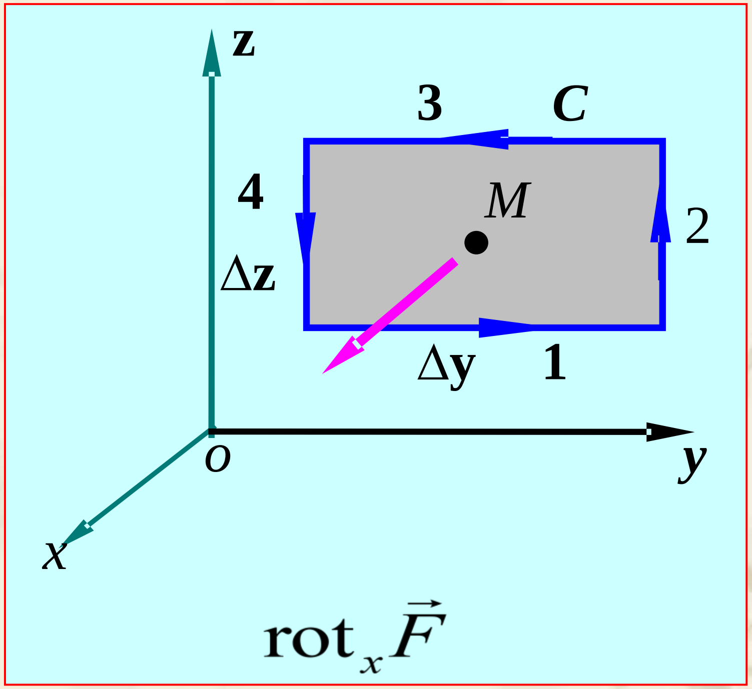
$$= F_{y1}\Delta y + F_{z2}\Delta z + F_{y3}(-\Delta y) + F_{z4}(-\Delta z)$$

$$F_{y1} \approx F_y(M) - \left.\frac{\partial F_y}{\partial z}\right|_M \frac{\Delta z}{2}$$

$$F_{z2} \approx F_z(M) + \left.\frac{\partial F_z}{\partial y}\right|_M \frac{\Delta y}{2}$$

$$F_{y3} \approx F_y(M) + \left.\frac{\partial F_y}{\partial z}\right|_M \frac{\Delta z}{2}$$

$$F_{z4} \approx F_z(M) - \left.\frac{\partial F_z}{\partial y}\right|_M \frac{\Delta y}{2}$$



Then $$\oint_C \vec{F}\cdot \mathrm{d}\vec{l} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\Delta y \Delta z$$

So $$\mathrm{rot}_x \vec{F} = \lim_{\Delta S\to 0} \frac{\oint_C \vec{F}\cdot \mathrm{d}\vec{l}}{\Delta S} = \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}$$

and $$\mathrm{rot}_y \vec{F} = \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \qquad \mathrm{rot}_z \vec{F} = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}$$

$$rot\ \vec{F} = \vec{e}_x\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \vec{e}_y\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \vec{e}_z\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$= \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$\boxed{\mathrm{rot}\ \vec{F} = \nabla \times \vec{F}}$$

The calculation formula of curl:

- **Cartesian Coordinates:**

$$\nabla \times \vec{F} = \vec{e}_x\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \vec{e}_y\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \vec{e}_z\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$= \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

- **Cylindrical Coordinates:**

$$\nabla \times \vec{F} = \frac{1}{\rho}\begin{vmatrix} \vec{e}_\rho & \rho\vec{e}_\phi & \vec{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_\rho & \rho F_\phi & F_z \end{vmatrix}$$

- **Spherical Coordinates:**

$$\nabla \times \vec{F} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \vec{e}_r & r\vec{e}_\theta & r\sin\theta\vec{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & rF_\theta & r\sin\theta F_\phi \end{vmatrix}$$

Exercise：

$$\vec{A} = \vec{e}_x 2 + \vec{e}_y 3 + \vec{e}_z 4$$

$$\nabla \times \vec{A} =$$

$$\vec{A} = \vec{e}_x 2x + \vec{e}_y 3y + \vec{e}_z 4z^2$$

$$\nabla \times \vec{A} =$$

$$\vec{A} = \vec{e}_x 2z^2 + \vec{e}_y 3x$$

$$\nabla \times \vec{A} =$$

The relevant formula of curl：

The divergence of a vector's Curl is zero

The curl of a scalar's gradient is zero

$$\left\{\begin{aligned}
&\nabla\times\vec{C}=0\\
&\nabla\times(f\vec{C})=\nabla f\times\vec{C}\\
&\nabla\times(f\vec{F})=f\nabla\times\vec{F}+\nabla f\times\vec{F}\\
&\nabla\times(\vec{F}\pm\vec{G})=\nabla\times\vec{F}\pm\nabla\times\vec{G}\\
&\nabla\cdot(\vec{F}\times\vec{G})=\vec{G}\cdot\nabla\times\vec{F}-\vec{F}\cdot\nabla\times\vec{G}\\
&\nabla\cdot(\nabla\times\vec{F})\equiv 0\\
&\nabla\times(\nabla u)\equiv 0
\end{aligned}\right.$$

$$\nabla \times \nabla u \equiv 0$$

Prove： left=$(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z) \times \left(\frac{\partial u}{\partial x}\vec{e}_x + \frac{\partial u}{\partial y}\vec{e}_y + \frac{\partial u}{\partial z}\vec{e}_z \right)$

$$= [(\frac{\partial^2 u}{\partial y \partial z} - \frac{\partial^2 u}{\partial z \partial y})\vec{e}_x + (\frac{\partial^2 u}{\partial z \partial x} - \frac{\partial^2 u}{\partial x \partial z})\vec{e}_y + (\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y \partial x})\vec{e}_z] = 0$$

$$\nabla\bullet(\nabla\times\vec{F})\equiv 0$$

$$\text{left}=\left(\frac{\partial}{\partial x}\vec{e}_x+\frac{\partial}{\partial y}\vec{e}_y+\frac{\partial}{\partial z}\vec{e}_z\right)\bullet$$

$$\left[\left(\frac{\partial F_z}{\partial y}-\frac{\partial F_y}{\partial z}\right)\vec{e}_x+\left(\frac{\partial F_x}{\partial z}-\frac{\partial F_z}{\partial x}\right)\vec{e}_y+\left(\frac{\partial F_y}{\partial x}-\frac{\partial F_x}{\partial y}\right)\vec{e}_z\right]$$

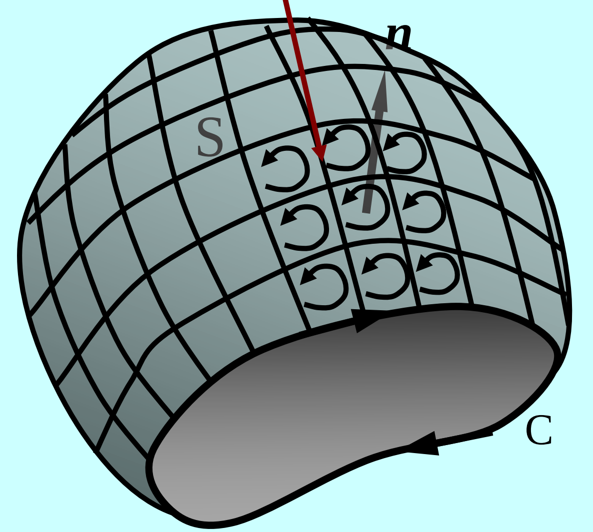
$$=\left[\left(\frac{\partial^2 F_z}{\partial x\partial y}-\frac{\partial^2 F_y}{\partial x\partial z}\right)+\left(\frac{\partial^2 F_x}{\partial y\partial z}-\frac{\partial^2 F_z}{\partial x\partial y}\right)+\left(\frac{\partial^2 F_y}{\partial x\partial z}-\frac{\partial^2 F_x}{\partial y\partial z}\right)\right]=0$$

3. Stokes's Theorem

The relationship between the circulation and rotation:

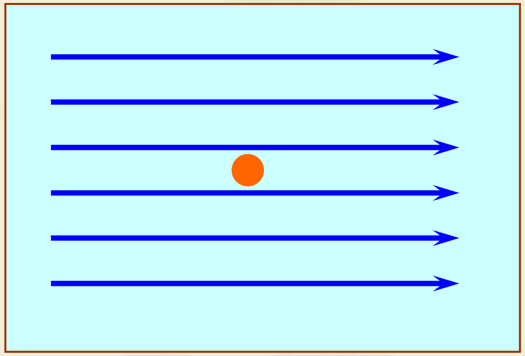
$$\oint_C \vec{F} \cdot \mathrm{d}\vec{l} = \int_S \nabla \times \vec{F} \cdot \mathrm{d}\vec{S}$$

Equally results in the opposite direction, will be cancelled by each other.

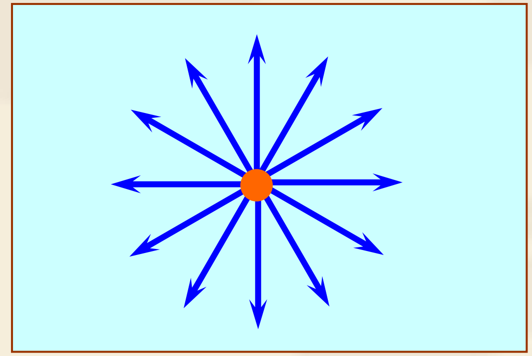


Surface subdivision

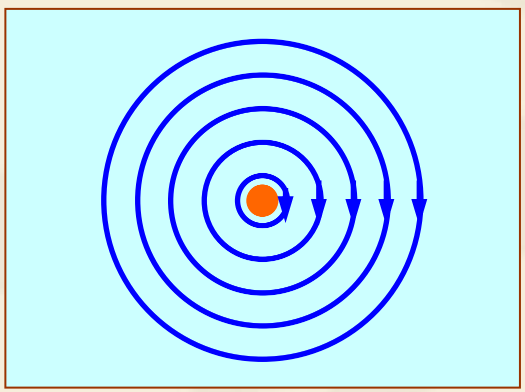
4. The difference of divergence and curl



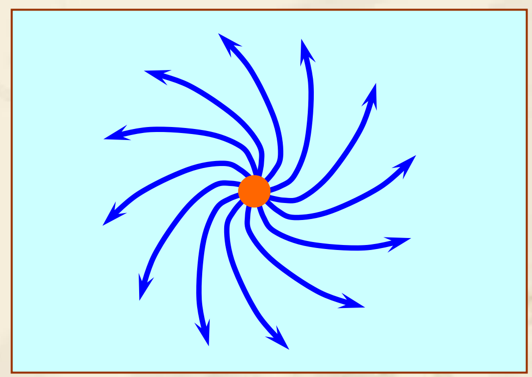
$$\nabla\cdot\vec{F}=0,\nabla\times\vec{F}=0$$



$$\nabla\cdot\vec{F}\neq 0.\nabla\times\vec{F}=0$$

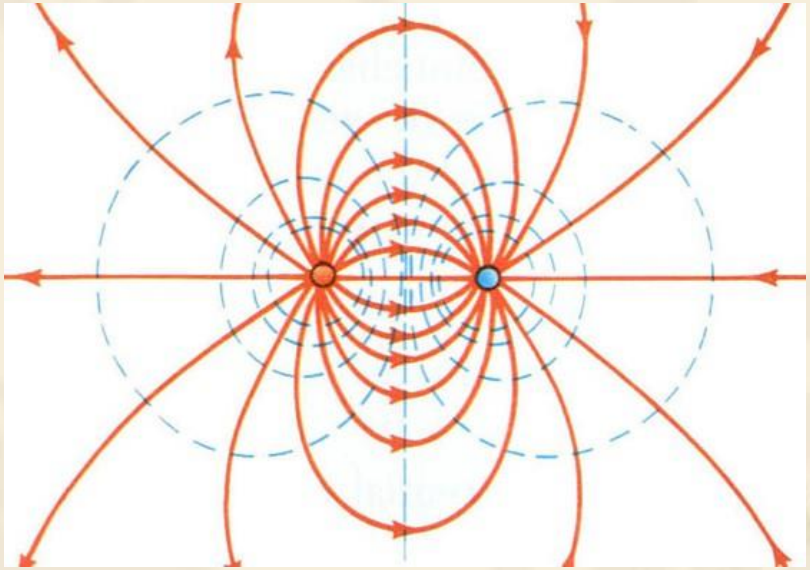


$$\nabla\cdot\vec{F}=0,\nabla\times\vec{F}\neq 0$$



$$\nabla\cdot\vec{F}\neq 0,\nabla\times\vec{F}\neq 0$$

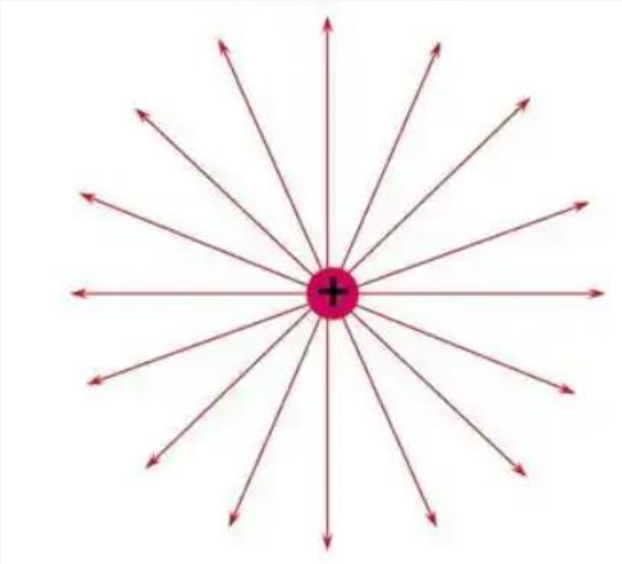
$$\nabla \cdot \vec{F} = 0? \qquad or \qquad \nabla \times \vec{F} = 0?$$



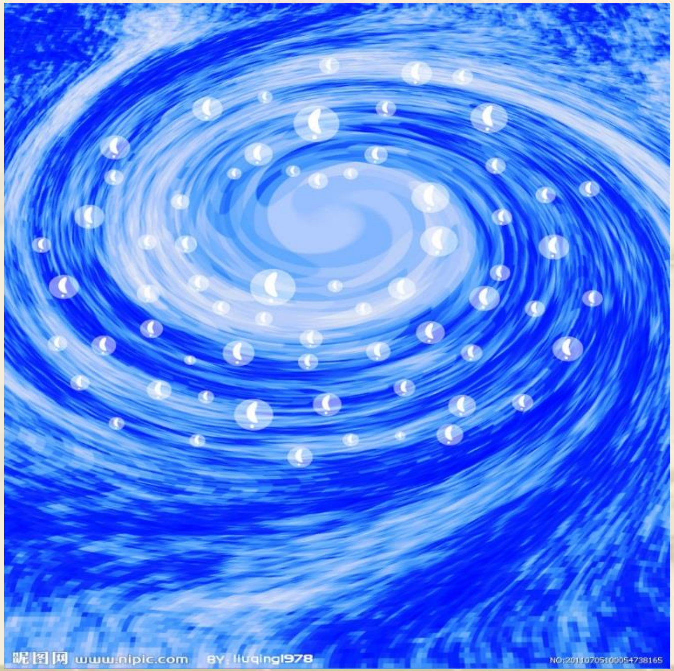
2.6 Irrotational field and solenoidal field

1. Sources of vector field

Divergence source



Vortex source



2. Classification of the vector field according to the field sources

(1) Irrotational field

Vector field has only divergence source without curl source: $\nabla\times\vec{F}\equiv 0$

$\oint_C \vec{F}\cdot \mathrm{d}\vec{l}=0$, **line integral independent with path, a conservative field。**

Irrotational field can be expressed using the gradient of a scalar field:

$$\vec{F}=-\nabla u$$

$$\boxed{\nabla\times\vec{F}=-\nabla\times(\nabla u)\equiv 0}$$

$$\int_S\left[\nabla\times(\nabla u)\right]\cdot d\vec{s} \overset{Stockes's\quad Theorem}{=} \oint_C(\nabla u)\cdot d\vec{l}=0$$

Example: Electrostatic field $\nabla\times\vec{E}\equiv 0\Rightarrow \vec{E}=-\nabla\varphi$

（2）Solenoidal field

Vector field has only curl source without divergence source:

Feature: $\oint_S \vec{F} \cdot \mathrm{d}\vec{S} = 0 \qquad \nabla \cdot \vec{F} \equiv 0$

Solenoidal field: can be expressed as the curl of another vector field

$$\vec{F} = \nabla \times \vec{A}$$

$$\boxed{\nabla \cdot \vec{F} = \nabla \cdot (\nabla \times \vec{A}) \equiv 0}$$

$$\int_V \left[\nabla \cdot (\nabla \times \mathbf{A})\right] dv \overset{\substack{\text{Divergence}\\\text{Theorem}}}{=} \oint_S (\nabla \times \mathbf{A}) \cdot ds = \int_{S_1} (\nabla \times \mathbf{A}) \cdot \mathbf{a}_{n1} ds + \int_{S_2} (\nabla \times \mathbf{A}) \cdot \mathbf{a}_{n2} ds$$

$$= \oint_{C_1} \mathbf{A} \cdot dl + \oint_{C_2} \mathbf{A} \cdot dl = 0$$

Example: **Static magnetic field**

$$\nabla \cdot \vec{B} = 0 \Rightarrow \vec{B} = \nabla \times \vec{A}$$

（3）**Irrotational and solenoidal field** **(Source is outside the discussed region)**

$$\nabla \times \vec{F} = 0 \Rightarrow \vec{F} = -\nabla u$$

$$\nabla \cdot \vec{F} = 0 \Rightarrow \nabla \cdot (-\nabla u) = 0$$

$$\Downarrow$$

$$\nabla^2 u = 0$$

（4）**Normal field neither solenoidal nor irrotational**

This field can be decomposed into two parts:
irrotational part and solenoidal part;

$$\vec{F}(\vec{r}) = \vec{F}_l(\vec{r}) + \vec{F}_C(\vec{r}) = -\nabla u(\vec{r}) + \nabla \times \vec{A}(\vec{r})$$

Irrotational part

Solenoidal part

2.7 Laplacian operation

- **Scalar laplacian operation** $\nabla^2 u$

$$\nabla \cdot (\nabla u) = \nabla^2 u \qquad \nabla^2$$ —— **Laplasse operation**

Equation：

Cartesian Coordinates:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

Cylindrical Coordinates:

$$\nabla^2 u = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho \frac{\partial u}{\partial \rho}) + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

Spherical Coordinates:

$$\nabla^2 u = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \frac{\partial u}{\partial r}) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}(\sin\theta \frac{\partial u}{\partial \theta}) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 u}{\partial \phi^2}$$

2.8 Helmholtz's Theorem

Helmholtz's Theorem:

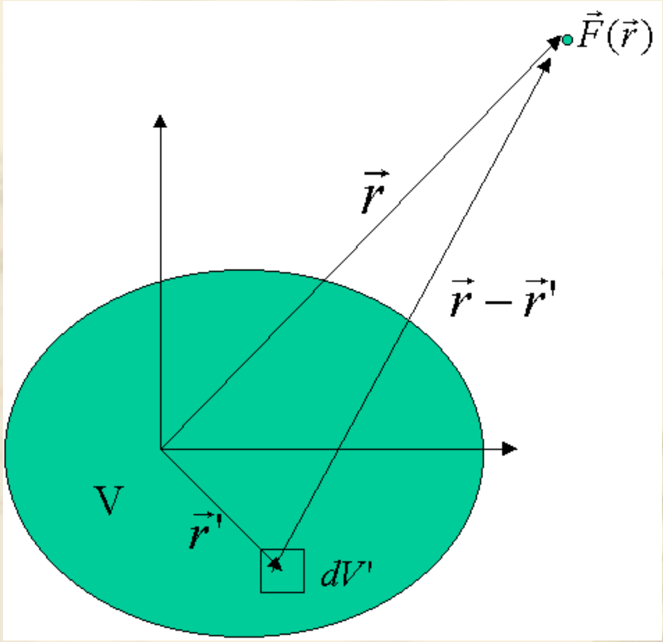
In an unbounded region, a vector field is determined to within an additive constant if both its divergence and its curl are specified everywhere：

$$\vec{F}(\vec{r}) = -\nabla u(\vec{r}) + \nabla \times \vec{A}(\vec{r})$$

Where :

$$u(\vec{r}) = \frac{1}{4\pi}\int_V \frac{\nabla' \cdot \vec{F}(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|}\mathrm{d}V'$$

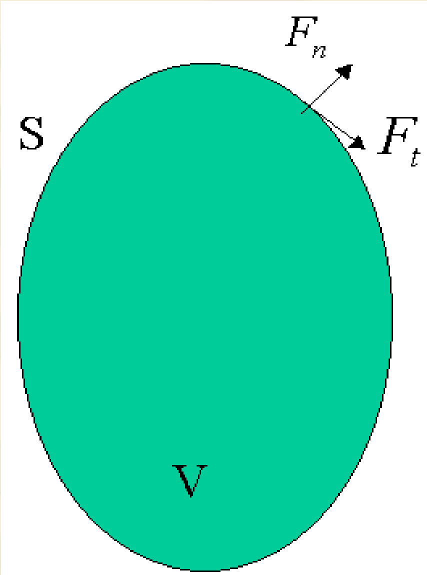
$$\vec{A}(\vec{r}) = \frac{1}{4\pi}\int_V \frac{\nabla' \times \vec{F}(\vec{r}\,')}{|\vec{r} - \vec{r}\,'|}\mathrm{d}V'$$



Vector field confined within a region bounded by a surface, is determined if its divergence and curl throughout the region, as well as the normal component of the vector over the bounding face, are given.

$$u(\vec{r}) = \frac{1}{4\pi}\int_V \frac{\nabla' \cdot \vec{F}(\vec{r}')}{|\vec{r}-\vec{r}'|}\mathrm{d}V' - \frac{1}{4\pi}\oint_S \frac{\vec{F}(\vec{r}')\cdot \mathrm{d}\vec{S}'}{|\vec{r}-\vec{r}'|}$$

$$\vec{A}(\vec{r}) = \frac{1}{4\pi}\int_V \frac{\nabla' \times \vec{F}(\vec{r}')}{|\vec{r}-\vec{r}'|}\mathrm{d}V' - \frac{1}{4\pi}\oint_S \frac{\vec{F}(\vec{r}')\times \mathrm{d}\vec{S}'}{|\vec{r}-\vec{r}'|}$$



Bounded area

Homework

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